

The vector and Banach lattices library

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Chapter 1

The Basics

This chapter develops the basic order-theoretic algebra that underpins the rest of the library: lattice-ordered abelian groups and the identities governing positive part, negative part and absolute value; vector lattices and their interaction with scalar multiplication; the Archimedean property; the Riesz decomposition theorem; normed and Banach lattices and the automatic continuity of the lattice operations; the calculus of disjointness; and finally the two order-completeness conditions (Dedekind and sigma Dedekind) together with their restriction to the positive cone.

1.1 Vector lattices and the Archimedean property

This section sets up the basic order-theoretic algebra underpinning the rest of the library. We work first in the generality of a lattice-ordered abelian group, establishing the fundamental identities for the positive part x^+ , the negative part x^- , and the absolute value $|x|$, together with the uniqueness of the positive part and the interaction of the lattice operations with translation and with arbitrary suprema and infima. We then specialise to *vector lattices*, i.e. real vector spaces with a compatible lattice structure, and record how non-negative scalars distribute over the lattice operations, culminating in the identity $|\lambda x| = |\lambda| |x|$. The section closes with the Archimedean property, which plays a foundational role in later chapters: we give two equivalent formulations and use them to show that an Archimedean vector lattice of dimension strictly greater than 1 necessarily contains vectors that are neither non-negative nor non-positive.

1.1.1 Lattice-ordered abelian groups

Throughout this subsection X denotes a *lattice-ordered abelian group*: an additive abelian group equipped with a lattice order whose translations are order-preserving. We write $x^+ := x \vee 0$, $x^- := (-x) \vee 0$ and $|x| := x \vee (-x)$ for the positive part, negative part and absolute value of $x \in X$.

Lemma 1.1.1. *Let X be a lattice-ordered abelian group and let $x \in X$. Then $|x| = 0$ if and only if $x = 0$.*

Lemma 1.1.2. *Let X be a lattice-ordered abelian group, let $x \in X$, and suppose $u, v \in X$ satisfy $x = u - v$ and $u \wedge v = 0$. Then $u = x^+$.*

Lemma 1.1.3. *For all x, y in a lattice-ordered abelian group, $x \vee y = x + (y - x)^+$.*

Lemma 1.1.4. *For all x, y in a lattice-ordered abelian group, $x \wedge y = x - (x - y)^+$.*

Lemma 1.1.5. For all x, y in a lattice-ordered abelian group, $x - x \wedge y = (x - y)^+$.

Lemma 1.1.6. For all x, y in a lattice-ordered abelian group, $(x + y)^+ \leq x^+ + y^+$.

Lemma 1.1.7. Let X be a lattice-ordered abelian group and let $x, a, b \in X$ with $0 \leq x$, $0 \leq a$, and $0 \leq b$. Then $x \wedge (a + b) \leq x \wedge a + x \wedge b$.

Lemma 1.1.8. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x + a = \sup\{x + z : z \in A\}$.

Lemma 1.1.9. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x + a = \inf\{x + z : z \in A\}$.

Lemma 1.1.10. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x \wedge a = \sup\{x \wedge z : z \in A\}$.

Lemma 1.1.11. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x \vee a = \inf\{x \vee z : z \in A\}$.

Lemma 1.1.12. Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a + b = \inf\{p + q : p \in A, q \in B\}$.

Lemma 1.1.13. Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \vee b = \inf\{p \vee q : p \in A, q \in B\}$.

Lemma 1.1.14. Let X be a lattice, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \wedge b = \inf(A \cup B)$.

1.1.2 Vector lattices

A *vector lattice* is a lattice-ordered abelian group equipped with a real vector space structure whose scalar multiplication by non-negative reals is monotone. Throughout this subsection X denotes a vector lattice.

Definition 1.1.15. Let X be an additive abelian group equipped with a lattice structure compatible with addition. A *vector lattice* structure on X consists of a real vector space structure on X for which scalar multiplication by non-negative reals is monotone: for all $\lambda \geq 0$ and $x \leq y$ in X , $\lambda x \leq \lambda y$.

Lemma 1.1.16. Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \vee y) = (\lambda x) \vee (\lambda y)$.

Lemma 1.1.17. Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \sup A$, then $\lambda a = \sup\{\lambda z : z \in A\}$.

Lemma 1.1.18. Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \inf A$, then $\lambda a = \inf\{\lambda z : z \in A\}$.

Lemma 1.1.19. Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \wedge y) = (\lambda x) \wedge (\lambda y)$.

Lemma 1.1.20. Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \vee \beta) \cdot x = (\alpha x) \vee (\beta x)$.

Lemma 1.1.21. Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \wedge \beta) \cdot x = (\alpha x) \wedge (\beta x)$.

Theorem 1.1.22. Let X be a vector lattice, let $x \in X$ and $\alpha \in \mathbb{R}$. Then $|\alpha x| = |\alpha| \cdot |x|$.

Lemma 1.1.23. Let X be a vector lattice, let $x, y \in X$ with $x \wedge y = 0$ and let $\lambda \in \mathbb{R}$ with $0 \leq \lambda$. Then $(\lambda x) \wedge y = 0$.

1.1.3 The Archimedean property

A lattice-ordered abelian group is *Archimedean* when no non-zero positive element is infinitesimal with respect to some other element. We record two equivalent formulations of this property and deduce that in an Archimedean vector lattice of dimension at least two there exist vectors which are neither negative nor positive.

Definition 1.1.24. A lattice-ordered abelian group X is called *Archimedean* when, for every $x, y \in X$ with $0 \leq x$ and $n \cdot x \leq y$ for all $n \in \mathbb{N}$, one has $x = 0$.

Lemma 1.1.25. Let X be a lattice-ordered abelian group. Then X is Archimedean if and only if for all $x, y \in X$ the condition $n \cdot x \leq y$ for all $n \in \mathbb{N}$ implies $x \leq 0$.

Lemma 1.1.26. A vector lattice X is Archimedean if and only if for every $u \in X$ with $0 \leq u$, $\inf\{n^{-1} \cdot u : n \in \mathbb{N}, n > 0\} = 0$.

Lemma 1.1.27. Let X be an Archimedean lattice-ordered abelian group and let $x, y \in X$. If $n \cdot |x| \leq y$ for all $n \in \mathbb{N}$, then $x = 0$.

Theorem 1.1.28. Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exists $x \in X$ which is neither negative nor positive.

1.2 Riesz decomposition

This section collects the Riesz decomposition theorem and its standard refinements in a lattice-ordered abelian group X . The binary form splits a positive element dominated by a sum of two positive elements into a corresponding sum. Iterating and combining the argument yields a matrix form for two partitions of the same positive sum, a version with a finite sum on the left hand side, a decomposition adapted to the absolute value, and an identification of sum intervals $[0, x] + [0, y] = [0, x + y]$ for $x, y \geq 0$.

Theorem 1.2.1. Let X be a lattice-ordered abelian group and let $x, y, z \in X$ with $0 \leq x$, $0 \leq y$, $0 \leq z$ and $x \leq y + z$. Then there exist $x_1, x_2 \in X$ with $0 \leq x_1 \leq y$, $0 \leq x_2 \leq z$ and $x = x_1 + x_2$.

Theorem 1.2.2. Let X be a lattice-ordered abelian group, let $m, n \in \mathbb{N}$, let $u : \{0, \dots, m-1\} \rightarrow X$ and $z : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq u_i$ for all i and $0 \leq z_j$ for all j , and $\sum_i u_i = \sum_j z_j$. Then there exists a family $x : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \rightarrow X$ with $0 \leq x_{ij}$ for all i, j such that $u_i = \sum_j x_{ij}$ for every i and $z_j = \sum_i x_{ij}$ for every j .

Theorem 1.2.3. Let X be a lattice-ordered abelian group, let $k \in \mathbb{N}$, let $u : \{0, \dots, k-1\} \rightarrow X$ and $x, y \in X$ with $0 \leq u_i$ for all i , $0 \leq x$, $0 \leq y$ and $\sum_i u_i \leq x + y$. Then there exist $v, w : \{0, \dots, k-1\} \rightarrow X$ with $0 \leq v_i$ and $0 \leq w_i$ for all i , $\sum_i v_i \leq x$, $\sum_i w_i \leq y$, and $u_i = v_i + w_i$ for every i .

Theorem 1.2.4. Let X be a lattice-ordered abelian group, let $x, u, v \in X$ with $0 \leq u$, $0 \leq v$ and $|x| = u + v$. Then there exist $y, z \in X$ with $x = y + z$, $|y| = u$ and $|z| = v$.

Theorem 1.2.5. Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$, let $x \in X$ and let $v : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq v_i$ for every i and $|x| \leq \sum_i v_i$. Then there exists $y : \{0, \dots, n-1\} \rightarrow X$ with $x = \sum_i y_i$ and $|y_i| \leq |x| \wedge v_i$ for every i .

Theorem 1.2.6. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. Then $[0, x] + [0, y] = [0, x + y]$ as subsets of X .

1.3 Normed and Banach lattices

A *normed vector lattice* is a real vector lattice equipped with a norm satisfying the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$. This compatibility between the norm and the lattice operations has a number of immediate consequences: the lattice operations $\vee$, $\wedge$ and $|\cdot|$ are norm-continuous, the space is automatically Archimedean, the positive cone is norm-closed, order intervals are norm-closed and norm-bounded, and monotone norm-convergent sequences converge to their supremum. A *Banach lattice* is a normed vector lattice whose norm is complete.

Definition 1.3.1. Let X be a normed abelian group equipped with a compatible lattice structure. A *normed vector lattice* structure on X consists of a vector lattice structure together with the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$ for all $x, y \in X$, and the identity $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$ and $x \in X$.

Lemma 1.3.2. Every normed vector lattice is a normed space over $\mathbb{R}$.

Theorem 1.3.3. Let X be a normed vector lattice. The map $(x, y) \mapsto x \vee y$ from $X \times X$ to X is continuous.

Theorem 1.3.4. Let X be a normed vector lattice. The map $(x, y) \mapsto x \wedge y$ from $X \times X$ to X is continuous.

Theorem 1.3.5. Let X be a normed vector lattice. The map $x \mapsto |x|$ from X to X is Lipschitz with constant 1.

Theorem 1.3.6. Every normed vector lattice is Archimedean.

Lemma 1.3.7. Let X be a normed vector lattice. The order relation on X is closed in the norm topology.

Theorem 1.3.8. Let X be a normed vector lattice. The positive cone $\{x \in X : 0 \leq x\}$ is norm-closed.

Theorem 1.3.9. Let X be a normed vector lattice, let $u, v : \mathbb{N} \rightarrow X$ and $a, b \in X$. If $u_n \rightarrow a$ and $v_n \rightarrow b$ in norm and $u_n \leq v_n$ for every n , then $a \leq b$.

Theorem 1.3.10. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be monotone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the least upper bound of the range of u .

Theorem 1.3.11. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be antitone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the greatest lower bound of the range of u .

Lemma 1.3.12. Let X be a normed vector lattice and let $a, b \in X$. The order interval $[a, b]$ is norm-closed.

Lemma 1.3.13. Let X be a normed vector lattice, let $a, b \in X$ and let $x \in [a, b]$. Then $\|x\| \leq \|a\| \vee \|b\|$.

Theorem 1.3.14. Let X be a normed vector lattice and let $s \subseteq X$ be both bounded below and bounded above in the order. Then s is norm-bounded.

Definition 1.3.15. A *Banach lattice* is a normed vector lattice whose underlying normed space is complete.

1.4 Disjointness

Two elements of a lattice-ordered abelian group are *disjoint* when the infimum of their absolute values vanishes. This section develops the elementary calculus of disjointness: symmetry, the zero rules, the Birkhoff identity $|x + y| = |x| + |y|$ for disjoint pairs, uniqueness of the positive/negative decomposition, compatibility with scalar multiplication and with monotonicity under the absolute value, and closure under finite suprema and sums. From these one deduces the finite-family identity $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i| |x_i|$ for pairwise-disjoint families, from which any pairwise-disjoint family of non-zero vectors is linearly independent over $\mathbb{R}$. The section closes with a normed-lattice statement: a norm limit of a pairwise-disjoint sequence must be zero.

Definition 1.4.1. Let X be a lattice-ordered abelian group and let $x, y \in X$. We say x and y are *disjoint* when $|x| \wedge |y| = 0$.

Lemma 1.4.2. Let X be a lattice-ordered abelian group and let $x, y \in X$. Then x and y are disjoint if and only if y and x are disjoint.

Lemma 1.4.3. Let X be a lattice-ordered abelian group and let $x \in X$. Then 0 and x are disjoint.

Lemma 1.4.4. Let X be a lattice-ordered abelian group and let $x \in X$. Then x and 0 are disjoint.

Lemma 1.4.5. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. If x and y are disjoint, then $x \wedge y = 0$.

Lemma 1.4.6. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $x \wedge y = 0$. Then x and y are disjoint.

Theorem 1.4.7. Let X be a lattice-ordered abelian group and let $u_1, v_1, u_2, v_2 \in X$ be non-negative elements. If u_1 and v_1 are disjoint, u_2 and v_2 are disjoint, and $u_1 - v_1 = u_2 - v_2$, then $u_1 = u_2$ and $v_1 = v_2$.

Theorem 1.4.8. Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| + |y|$.

Lemma 1.4.9. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $|x \vee y| = |x| \vee |y|$.

Lemma 1.4.10. Let X be a lattice-ordered abelian group and let $x \in X$. Then x^+ and x^- are disjoint.

Lemma 1.4.11. Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |y|$, then x and z are disjoint.

Lemma 1.4.12. Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |x|$, then z and y are disjoint.

Lemma 1.4.13. Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, αx and y are disjoint.

Lemma 1.4.14. Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, x and αy are disjoint.

Theorem 1.4.15. Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exist non-zero $u, v \in X$ with u and v disjoint.

Lemma 1.4.16. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $x + y = x \vee y$.*

Lemma 1.4.17. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x| \vee |y| = |x| + |y|$.*

Lemma 1.4.18. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| \vee |y|$.*

Lemma 1.4.19. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y \vee z$ are disjoint.*

Lemma 1.4.20. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y + z$ are disjoint.*

Lemma 1.4.21. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x \vee y$ and z are disjoint.*

Lemma 1.4.22. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x + y$ and z are disjoint.*

Lemma 1.4.23. *Let X be a lattice-ordered abelian group and let $x, y \in X$. Then $x - x \wedge y$ and $y - x \wedge y$ are disjoint.*

Theorem 1.4.24. *Let X be a vector lattice, let $x, y \in X$ and $\lambda \in \mathbb{R}$ with $0 < \lambda$. Then $(x - \lambda y)^+$ and $(y - \lambda^{-1}x)^+$ are disjoint.*

Theorem 1.4.25. *Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$ with $n > 0$, and let $f : \{0, \dots, n-1\} \rightarrow X$ be a family of non-negative elements which is pairwise disjoint. Then $\sum_i f_i = \bigvee_i f_i$.*

Theorem 1.4.26. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family. For every $\alpha : \{0, \dots, n-1\} \rightarrow \mathbb{R}$, $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i| |x_i|$.*

Theorem 1.4.27. *Let X be a vector lattice, let $n \in \mathbb{N}$ with $n > 0$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family of non-negative elements. For every non-negative family $\alpha : \{0, \dots, n-1\} \rightarrow \mathbb{R}$, $\sum_i \alpha_i x_i = \bigvee_i \alpha_i x_i$.*

Theorem 1.4.28. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family of non-zero elements. Then x is linearly independent over $\mathbb{R}$.*

Theorem 1.4.29. *Let Y be a normed vector lattice, let $u : \mathbb{N} \rightarrow Y$ be a pairwise-disjoint sequence, and let $x \in Y$. If $u_n \rightarrow x$ in norm, then $x = 0$.*

1.5 Order completeness

A lattice is *Dedekind complete* (or *order complete*) when every non-empty subset that is bounded above admits a supremum, and its *sigma* analogue is obtained by restricting the requirement to countable subsets. On a vector lattice either property can be checked on the restricted class of non-negative elements, and indeed only on increasing bounded above nets (respectively sequences) of non-negative elements. The section closes by showing that every Dedekind complete lattice is sigma Dedekind complete, and that sigma Dedekind completeness already forces the Archimedean property in any lattice-ordered abelian group.

On a vector lattice, Dedekind completeness can be checked either on increasing bounded above nets of non-negative elements, or on non-empty bounded above sets of non-negative elements.

Theorem 1.5.1. *Let X be a vector lattice. If every bounded above increasing net of non-negative elements of X admits a least upper bound, then X is Dedekind complete.*

Theorem 1.5.2. *Let X be a vector lattice. If every non-empty bounded above set of non-negative elements of X admits a least upper bound, then X is Dedekind complete.*

1.5.1 Sigma order completeness

A lattice is *sigma Dedekind complete* (or *sigma order complete*) when every non-empty countable subset that is bounded above admits a supremum, and every non-empty countable subset that is bounded below admits an infimum.

Definition 1.5.3. A lattice X is *sigma Dedekind complete* when every non-empty countable subset of X which is bounded above admits a supremum, and every non-empty countable subset of X which is bounded below admits an infimum.

Lemma 1.5.4. *Every Dedekind complete lattice is sigma Dedekind complete.*

As with Dedekind completeness, sigma Dedekind completeness on a vector lattice can be checked on the restricted class of non-negative elements, or even only on increasing bounded above sequences of non-negative elements.

Theorem 1.5.5. *Let X be a vector lattice. If every bounded above increasing sequence of non-negative elements of X admits a least upper bound, then X is sigma Dedekind complete.*

Theorem 1.5.6. *Let X be a vector lattice. If every non-empty bounded above countable set of non-negative elements of X admits a least upper bound, then X is sigma Dedekind complete.*

Sigma conditional completeness alone forces the Archimedean property in the lattice-ordered group setting.

Theorem 1.5.7. *Every sigma Dedekind complete lattice-ordered abelian group is Archimedean.*

Chapter 2

Substructures

2.1 Vector sublattices

A *vector sublattice* of a vector lattice X is a linear subspace closed under the lattice operations. Since the lattice operations are inter-definable, closure under any one of them is enough: closure under $\vee$ or under $|\cdot|$ both characterise the notion. This section introduces the bundled structure, shows that every vector sublattice is a vector lattice in its own right and that the collection of vector sublattices of a fixed ambient space forms a complete lattice under inclusion, and describes the vector sublattice generated by a set in several concrete situations — a submodule, a pointed cone, and a pairwise-disjoint family of non-negative vectors. The section closes with the normed setting: the induced normed vector lattice structure on the subtype of a vector sublattice, its topological closure, completeness when the ambient space is a Banach lattice, and separability of the closed vector sublattice generated by a countable subset.

2.1.1 Pointed cones

Before introducing vector sublattices themselves, we record that in a vector lattice the sup-closure and inf-closure of a pointed cone are again pointed cones. These two operations will appear as basic building blocks in the descriptions of the generated vector sublattice. Recall that a *pointed cone* is a subset of a vector subspace that is closed under addition, positive scalar multiplication, and contains the origin.

Lemma 2.1.1. *Let X be a vector lattice and let C be a pointed cone in X . Then the sup-closure of C , viewed as a subset of X , is a pointed cone.*

Lemma 2.1.2. *Let X be a vector lattice and let C be a pointed cone in X . Then the inf-closure of C , viewed as a subset of X , is a pointed cone.*

2.1.2 Vector sublattices

Throughout the rest of the section X denotes a vector lattice.

Definition 2.1.3. Let X be a vector lattice. A *vector sublattice* of X is a linear subspace $Y \subseteq X$ which is closed under $\vee$: for all $x, y \in Y$, $x \vee y \in Y$.

Lemma 2.1.4. *Let Y be a vector sublattice of X and let $x, y \in Y$. Then $x \wedge y \in Y$.*

Lemma 2.1.5. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^+ \in Y$.*

Lemma 2.1.6. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^- \in Y$.*

Lemma 2.1.7. *Let Y be a vector sublattice of X and let $x \in Y$. Then $|x| \in Y$.*

Lemma 2.1.8. *Every vector sublattice of X is a sublattice of the underlying lattice of X .*

Since the lattice operations of a vector lattice are inter-definable with the absolute value, a linear subspace closed under $|\cdot|$ is automatically a vector sublattice.

Lemma 2.1.9. *A linear subspace of X closed under $|\cdot|$ (i.e. $|x| \in M$ for every $x \in M$) is a vector sublattice of X .*

Lemma 2.1.10. *Let M be a linear subspace of X . Then M is closed under $|\cdot|$ if and only if M is closed under $\vee$.*

A vector sublattice inherits a vector lattice structure from the ambient space, with the lattice operations, addition and scalar multiplication computed pointwise.

Lemma 2.1.11. *Let Y be a vector sublattice of X . Then Y is itself a vector lattice, with lattice operations, addition, and scalar multiplication inherited pointwise from X .*

The collection of vector sublattices of X is ordered by inclusion. It has a greatest element $\top = X$, a least element $\perp = \{0\}$, and is closed under arbitrary intersections.

Lemma 2.1.12. *The forgetful map from vector sublattices of X to linear subspaces of X is injective.*

Lemma 2.1.13. *The whole space X is the greatest vector sublattice of X , the zero subspace $\{0\}$ is the least vector sublattice of X , and the intersection of an arbitrary family of vector sublattices is again a vector sublattice of X .*

2.1.3 The vector sublattice generated by a set

For any subset $s \subseteq X$, there is a smallest vector sublattice of X containing s , obtained as the infimum of all vector sublattices that contain s . The subsection characterises this generated sublattice in several concrete situations.

Definition 2.1.14. Let s be a subset of X . The vector sublattice *generated* by s is the intersection of all vector sublattices of X containing s .

Lemma 2.1.15. *For every subset $s \subseteq X$, s is contained in the vector sublattice generated by s .*

Lemma 2.1.16. *Let $s \subseteq X$ and let Y be a vector sublattice of X with $s \subseteq Y$. Then the vector sublattice generated by s is contained in Y .*

Theorem 2.1.17. *Let M be a linear subspace of X . Then the underlying set of the vector sublattice generated by M equals the sup-closure of the inf-closure of M :*

$$\text{generated}(M) = \text{supClosure}(\text{infClosure}(M)).$$

Theorem 2.1.18. *Let M be a linear subspace of X . Then the underlying set of the vector sublattice generated by M equals the inf-closure of the sup-closure of M :*

$$\text{generated}(M) = \text{infClosure}(\text{supClosure}(M)).$$

Theorem 2.1.19. *Let C be a pointed cone in X . Then the underlying set of the vector sublattice generated by C equals the difference set of the sup-closure of C with itself:*

$$\text{generated}(C) = \text{supClosure}(C) - \text{supClosure}(C).$$

Theorem 2.1.20. *Let C be a pointed cone in X . Then the underlying set of the vector sublattice generated by C equals the difference set of the inf-closure of C with itself:*

$$\text{generated}(C) = \text{infClosure}(C) - \text{infClosure}(C).$$

Theorem 2.1.21. *Let $A \subseteq X$ be a set of pairwise-disjoint elements with $0 \leq a$ for every $a \in A$. Then the underlying submodule of the vector sublattice generated by A coincides with the linear span of A over $\mathbb{R}$.*

2.1.4 Normed and Banach sublattices

In the normed setting, a vector sublattice Y of a normed vector lattice X inherits a normed vector lattice structure on its subtype. The topological closure of a vector sublattice is again a vector sublattice; in a Banach lattice, norm-closed vector sublattices are themselves Banach lattices. Finally, the closed vector sublattice generated by a countable subset is separable.

Lemma 2.1.22. *Let X be a normed vector lattice and let Y be a vector sublattice of X . Then Y is itself a normed vector lattice under the induced norm and lattice operations.*

Lemma 2.1.23. *Let X be a normed vector lattice and let Y be a vector sublattice of X . Then the norm closure of Y is a vector sublattice of X .*

Lemma 2.1.24. *Let X be a Banach lattice and let Y be a norm-closed vector sublattice of X . Then Y is a Banach lattice under the induced structures.*

Theorem 2.1.25. *Let X be a normed vector lattice and let $A \subseteq X$ be countable. Then the topological closure of the vector sublattice generated by A is separable.*

2.2 Ideals

An *order ideal* of a vector lattice X is a vector sublattice J of X which is *solid*: whenever $x \in J$ and $0 \leq y \leq x$, y lies in J as well. Equivalently, a linear subspace M of X is solid in the absolute-value sense $|y| \leq |x| \Rightarrow y \in M$ if and only if it is closed under $\vee$ and solid in the positive-cone sense; in either formulation the resulting notion coincides with that of a solid subspace. This section develops the basic theory of order ideals: it introduces the bundled structure, produces the principal ideal generated by a single vector and equips it, in the Archimedean setting, with a canonical gauge (or order-unit) norm that makes it a normed vector lattice, shows that the ideals of X ordered by inclusion form a complete lattice, produces a proper non-trivial ideal whenever the dimension of X is strictly greater than 1, and studies the topological closure of an ideal in a normed vector lattice together with the resulting notion of closed order ideal in a Banach lattice.

2.2.1 Order ideals

Throughout the rest of the section X denotes a vector lattice.

Definition 2.2.1. Let X be a vector lattice. An *order ideal* of X is a vector sublattice J of X which is *solid*: for all $x, y \in X$ with $x \in J$, $0 \leq y$ and $y \leq x$, $y \in J$.

Lemma 2.2.2. *Let J be an order ideal of X and let $x \in X$. If $|x| \in J$, then $x \in J$.*

Lemma 2.2.3. *Let J be an order ideal of X and let $x, y \in X$ with $x \in J$ and $|y| \leq |x|$. Then $y \in J$.*

Lemma 2.2.4. *A linear subspace of X which is solid in the absolute-value sense (for all $x, y \in X$ with $x \in M$ and $|y| \leq |x|$, $y \in M$) is an order ideal of X .*

Lemma 2.2.5. *Let M be a linear subspace of X . Then M is solid in the sense of the absolute value (for all $x, y \in X$ with $x \in M$ and $|y| \leq |x|$, $y \in M$) if and only if M is both solid in the positive-cone sense (for all $x, y \in X$ with $x \in M$, $0 \leq y$ and $y \leq x$, $y \in M$) and closed under $\vee$ (for all $x, y \in M$, $x \vee y \in M$).*

Lemma 2.2.6. *The forgetful map from order ideals of X to linear subspaces of X is injective.*

2.2.2 The principal ideal

Every vector $a \in X$ determines an order ideal I_a of X , the *principal ideal* of a , consisting of the vectors whose absolute value is dominated by a non-negative scalar multiple of $|a|$.

Definition 2.2.7. Let $a \in X$. The *principal ideal* generated by a is the order ideal I_a of X whose underlying set is

$$\{x \in X : \exists c \in \mathbb{R}, 0 \leq c \text{ and } |x| \leq c|a|\}.$$

Lemma 2.2.8. *Let $a \in X$. Then $a \in I_a$.*

2.2.3 The gauge norm on the principal ideal

Every vector in the principal ideal I_a carries a canonical non-negative quantity: the infimum of all non-negative scalars c for which $|x| \leq c|a|$. This is the *gauge norm* (or *order-unit norm*) of x with respect to a . The gauge norm is absolutely homogeneous on X , and subadditive and monotone in $|\cdot|$ on the principal ideal; together with its definiteness — which requires X to be Archimedean — this makes I_a a normed vector lattice whenever $a \neq 0$.

Definition 2.2.9. Let $a, x \in X$. The *gauge norm* of x with respect to a is

$$\|x\|_a := \inf\{c \in \mathbb{R} : 0 \leq c \text{ and } |x| \leq c|a|\},$$

with the convention that the infimum of an empty subset of $\mathbb{R}$ is 0.

Lemma 2.2.10. *Let $a, x \in X$. Then $0 \leq \|x\|_a$.*

Lemma 2.2.11. *Let $a \in X$. Then $\|0\|_a = 0$.*

Lemma 2.2.12. *Let $a, x \in X$. Then $\|-x\|_a = \|x\|_a$.*

Lemma 2.2.13. *Let $a \in X$ and let $x, y \in I_a$. Then $\|x + y\|_a \leq \|x\|_a + \|y\|_a$.*

Lemma 2.2.14. *Let $a, x \in X$ and let $r \in \mathbb{R}$. Then $\|r \cdot x\|_a = |r| \|x\|_a$.*

Theorem 2.2.15. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $|x| \leq \|x\|_a \cdot |a|$.*

Lemma 2.2.16. *Let $a, x \in X$ and let $c \in \mathbb{R}$ with $0 \leq c$ and $|x| \leq c|a|$. Then $\|x\|_a \leq c$.*

Lemma 2.2.17. *Let $a \in X$, let $y \in I_a$ and let $x \in X$ with $|x| \leq |y|$. Then $\|x\|_a \leq \|y\|_a$.*

Lemma 2.2.18. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $\|x\|_a = 0$ if and only if $x = 0$.*

Lemma 2.2.19. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $\|x\|_a \leq 1$ if and only if $|x| \leq |a|$.*

Lemma 2.2.20. *Let $e, u \in X$ with $0 \leq e$ and $e \leq u$, and let $x \in I_e$. Then $\|x\|_u \leq \|x\|_e$.*

Theorem 2.2.21. *Let X be an Archimedean vector lattice and let $a \in X$ with $a \neq 0$. Then I_a , equipped with the gauge norm $\|\cdot\|_a$ and with the lattice and vector-space operations inherited pointwise from X , is a normed vector lattice.*

2.2.4 Sum of ideals

The sum $J_1 + J_2$ of two order ideals, viewed as linear subspaces of X , is itself again an order ideal: every element of the sum splits into pieces which remain bounded in absolute value by the element itself. This yields two decomposition statements, one for non-negative elements and one with lattice estimates on the absolute values of the pieces.

Lemma 2.2.22. *Let J_1, J_2 be order ideals of X . Their sum $J_1 + J_2$, viewed as a linear subspace of X , is again an order ideal of X .*

Theorem 2.2.23. *Let J_1, J_2 be order ideals of X and let $y \in J_1 + J_2$ with $0 \leq y$. Then there exist $y_1, y_2 \in X$ with $y_1 \in J_1$, $y_2 \in J_2$, $y_1 + y_2 = y$, $0 \leq y_1$ and $0 \leq y_2$.*

Theorem 2.2.24. *Let J_1, J_2 be order ideals of X and let $z \in J_1 + J_2$. Then there exist $a, b \in X$ with $a \in J_1$, $b \in J_2$, $a + b = z$, $|a| \leq |z|$ and $|b| \leq |z|$.*

2.2.5 The complete lattice of order ideals

Ordered by inclusion, order ideals of X form a complete lattice: the trivial subspace $\{0\}$ is the least ideal, the whole space is the greatest, arbitrary intersections are again ideals, and binary joins are given by the sum. When X is Archimedean with dimension strictly greater than 1, this lattice always contains an ideal distinct from both $\{0\}$ and X .

Lemma 2.2.25. *The whole space X is the greatest order ideal of X , the zero subspace $\{0\}$ is the least order ideal of X , the intersection of an arbitrary family of order ideals is again an order ideal, and the order ideals of X ordered by inclusion form a complete lattice with binary joins given by the sum of ideals.*

Theorem 2.2.26. *Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exists an order ideal J of X with $J \neq \{0\}$ and $J \neq X$.*

2.2.6 Closure of an ideal

In a normed vector lattice, the topological closure of an order ideal is again an order ideal: the solid condition propagates to the closure by truncating an approximating sequence against the ambient absolute value. In a Banach lattice this closure operation gives rise to the bundled notion of closed order ideal, stable under arbitrary intersections and under binary joins given by the sum.

Lemma 2.2.27. *Let X be a normed vector lattice, let J be an order ideal of X , and let $x, y \in X$ with x in the topological closure of J and $|y| \leq |x|$. Then y lies in the topological closure of J .*

Lemma 2.2.28. *Let X be a normed vector lattice and let J be an order ideal of X . Then the topological closure of J is an order ideal of X .*

Definition 2.2.29. Let X be a Banach lattice. A *closed order ideal* of X is an order ideal of X whose underlying set is closed in the norm topology.

Lemma 2.2.30. *Let X be a Banach lattice. The intersection of two closed order ideals of X is a closed order ideal of X .*

Lemma 2.2.31. *Let X be a Banach lattice and let J_1, J_2 be closed order ideals of X . Then the sum $J_1 + J_2$ is a closed order ideal of X .*

Lemma 2.2.32. *Let X be a Banach lattice. The intersection of an arbitrary family of closed order ideals of X is a closed order ideal of X .*

Lemma 2.2.33. *Let X be a Banach lattice. The closed order ideals of X ordered by inclusion form a lattice with binary joins given by the sum and binary meets by intersection, and are closed under arbitrary intersections.*